

FitBuddy – AI Fitness Plan Generator using Gemini Models

Project Submission Report

■ Team Details

- **Team Leader:** Mohammed Razik Rizwan N
- **Team Member:** Muthu Raj S
- **Team Member:** Narayn T

■ Project Description

FitBuddy is a web-based application that uses artificial intelligence to generate personalized workout plans and nutrition tips based on user fitness goals, such as weight loss, muscle gain, or general wellness. Built using the **FastAPI framework** and integrated with Google's **Gemini AI models**, FitBuddy simplifies the process of creating structured, dynamic, and effective fitness routines. The application captures essential user parameters—such as age, weight, goals, and preferred workout intensity—to automatically construct a customized 7-day fitness regimen. Users can dynamically update and refine their routines through an interactive AI feedback loop, accompanied by targeted nutrition and recovery tips to maximize real-world results. Supported by a persistent SQLite database, the platform offers an intuitive frontend interface along with interactive developer documentation.

■ Operational Scenarios

- **Scenario 1 (Personalized 7-Day Workout Generation):** A user inputs personal information (name, age, weight, fitness goal, and intensity preference). The system instantly produces an organized, day-by-day training timeline customized to their target profile.
- **Scenario 2 (Feedback-Driven Adaptation):** Users can submit modification notes (e.g., *"more focus on cardio"* or *"include more rest days"*). The system safely captures the text, updates the existing record via the AI layer, and returns a newly tailored structure.
- **Scenario 3 (Goal-Aligned Nutrition & Recovery Tips):** The platform generates concise and practical diet or recovery suggestions (e.g., *"Include protein in your post-workout meal"*) that directly complement the physical training strategy.
- **Scenario 4 (Administrative Oversight Dashboard):** Coaches or administrators gain full access to a centralized dashboard showing all registered active users, their demographic data, and a side-by-side comparison of original versus updated fitness plans.

■ ■ Technical Architecture & Directory Blueprint

The software utilizes a modular, persistent backend structure to handle programmatic calls to Google Generative AI:

```
fitbuddy/
  ■ ■ ■ requirements.txt          # Project dependencies (FastAPI, SQLAlchemy, etc.)
  ■ ■ ■ app/
    ■ ■ ■ main.py                # FastAPI system entry point
    ■ ■ ■ routes.py              # Core application routing & request handlers
    ■ ■ ■ database.py            # SQLite configuration & SQLAlchemy ORM model logic
    ■ ■ ■ schemas.py             # Pydantic validation structures
    ■ ■ ■ gemini_generator.py     # Gemini 1.5 Pro integration for 7-day workout plans
    ■ ■ ■ gemini_flash_generator.py # Gemini Flash integration for practical nutrition tips
    ■ ■ ■ updated_plan.py        # Secondary logic engine handling feedback adjustments
    ■ ■ ■ nutrition.py           # Supplementary structural logic files
  ■ ■ ■ templates/
    ■ ■ ■ index.html             # Frontend user registration form
    ■ ■ ■ result.html            # Dynamic display page (Workout plan, tips, feedback)
    ■ ■ ■ all_users.html         # Administrative metrics panel
  ■ ■ ■ static/
    ■ ■ ■ images/
      ■ ■ ■ gym-bg.jpg          # Gym-themed visual styling assets
  ■ ■ ■ fitbuddy.db              # Local persistent SQLite database instance
```

Target Milestone Breakdown & Implementation Details

Milestone 1: Model Selection & Architecture Setup

- **Model Evaluation:** Hugging Face models like T5, BART, and Bloom were evaluated but rejected due to high hosting costs, local GPU requirements, and complex prompting overhead to produce structured text.
- **Final AI Selection: Gemini 1.5 Pro** was chosen for core workout scheduling and feedback alterations because of its rich contextual processing. **Gemini Flash** was selected to power the nutrition layer to ensure lightweight, fast API response times.
- **Database Infrastructure:** Set up **SQLite** along with **SQLAlchemy ORM** containing two core tables: a *users* table for metadata profiles and a *plans* table to track original and updated text outputs.

Milestone 2 & 3: Core Functionality & Backend Routing

- **Workout Pipeline (gemini_generator.py):** Packages user criteria into structured prompts specifying warm-ups (5–10 mins), targeted main exercise sets/reps, and safe cooldown steps.
- **Feedback Loop (updated_plan.py):** Feeds the active *original_plan* and raw *user_feedback* string to the AI model, safely modifying the requested segments while preserving the remaining layout.
- **FastAPI Routing Layer (routes.py):** Implements dynamic endpoints such as */generate-workout/gemini* and */update-plan/{user_id}* to securely handle user forms, interact with the DB, and render Jinja2 context variables.

Milestone 4 & 5: Frontend Design & Local Deployment

- **UI Implementation:** Created responsive, accessible HTML forms styled via embedded flexbox grid layouts with dark fitness-themed designs.
- **Dynamic Data Binding:** Integrated Jinja2 statements to securely pass background datasets into clean frontend user panels and tabular admin reports.
- **Local Production Verification:** Configured server execution parameters through Uvicorn using hot-reloading configurations, mapping explicit debugging utilities to *http://127.0.0.1:8000/docs*.

■ Conclusion

The **FitBuddy** engine successfully showcases how modern Generative AI models can enhance user engagement in health-tech ecosystems. By separating workloads across dedicated models (**Gemini 1.5 Pro** for contextual plan generation and **Gemini Flash** for fast utility responses), the development team created an efficient, scalable, and responsive application. This design establishes an effective blueprint for personalized wellness software.

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